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DiffusionGemma Technical Report

Google DeepMind’s Open 26B MoE Model Generates Text at 1000+ Tokens per Second via Block-Parallel Discrete Diffusion

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Autoregressive text generation has a fundamental throughput ceiling: it emits one token per forward pass, making latency proportional to sequence length. *DiffusionGemma* eliminates this bottleneck by iteratively denoising an entire 256-token canvas in parallel, achieving over 1000 tokens per second on a single NVIDIA H100—up to 4× faster than comparable AR models—while fitting within 18 GB VRAM when quantised and released under the Apache 2.0 licence.

AT A GLANCE

Problem: Autoregressive LLMs generate one token per forward pass, making throughput fundamentally linear in sequence length and bottlenecked by memory bandwidth.

Why it matters: A 4× throughput gain at matched quality halves the inference cost of frontier LLMs, directly enabling larger, cheaper deployments.

Method: Fine-tune the Gemma 4 26B MoE backbone with (1) SFT for bidirectional denoising and (2) RL + sampler distillation for joint quality and efficiency gains.

Result: 1000+ tok/s on H100, 700+ tok/s on RTX 5090; fits in 18 GB VRAM when quantised; open-weights under Apache 2.0.

Evidence: Speed benchmarks on NVIDIA H100 and RTX 5090; quality evaluated against equivalent-parameter AR models.

The Big Picture

Large language models generate text by sampling one token at a time in a strict left-to-right order. This autoregressive (AR) constraint is simple and ensures the model can condition each new token on all prior context, but it imposes a hard throughput ceiling: no matter how wide the model is, each forward pass produces exactly one token. For a sequence of N tokens, the model must run N serial passes, making decode time $O(N)$ in the number

of forward passes regardless of hardware parallelism.

Diffusion-based text generation breaks this constraint by treating generation as iterative denoising over a complete sequence. DiffusionGemma represents the state of the art on this front: a 26B-parameter open-weights model that generates full blocks of tokens in parallel, with measured throughput exceeding 1000 tokens per second on a single H100—a 4× improvement over equivalent AR models.

Why Existing Approaches Fall Short

Earlier attempts at diffusion-based language models (e.g. MDLM, SEDD) trained from scratch on text corpora and did not match the quality of AR models of similar parameter count. The quality gap arose because diffusion LMs require learning bidirectional context modelling, a fundamentally harder pre-training problem than the unidirectional next-token prediction that gives AR models their strong priors.

Prior fast-generation methods for AR models (speculative decoding, token streaming) provide modest speedups (2–3×) but remain bounded by the sequential auto-regressive paradigm. DiffusionGemma bypasses this bound entirely.

Core Mechanism

Architecture. DiffusionGemma is based on Gemma 4, specifically the 26B-A4B Mixture-of-Experts variant: 26B total parameters with only 3.8B activated per token during inference. The sparse MoE routing means most expert parameters are dormant in any given forward pass, keeping latency low while retaining the model’s large capacity.

Block-Autoregressive Multi-Canvas Sampling. At inference, DiffusionGemma begins with a canvas of 256 [MASK] tokens. Each denoising iteration:

1. Predicts the probability distribution over the full vocabulary for every masked position simultaneously;
2. Samples and commits the most confident subset of positions;
3. Repeats until the canvas is fully populated.

Once committed, positions are not revised. In practice far fewer than 256 iterations are required, as many positions are filled confidently in the first few steps.

Technical Formulation

Let $\mathbf{x} = (x_1, \dots, x_L)$ be a sequence of L tokens. At each denoising step t the model receives a partially masked sequence $\tilde{\mathbf{x}}^{(t)}$ and predicts a distribution over unmasked completions:

$$p_{\theta}(\mathbf{x} \mid \tilde{\mathbf{x}}^{(t)}) = \prod_{i \in \mathcal{M}^{(t)}} p_{\theta}(x_i \mid \tilde{\mathbf{x}}^{(t)}),$$

where $\mathcal{M}^{(t)}$ is the set of masked positions at step t . The SFT objective is:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{\mathbf{x}, t, \mathcal{M}} \left[\sum_{i \in \mathcal{M}} \log p_{\theta}(x_i \mid \tilde{\mathbf{x}}^{(t)}) \right].$$

The RL stage optimises a reward $r(\mathbf{x})$ that combines generation quality (from a judge model) and inference efficiency (number of denoising steps to convergence):

$$\mathcal{L}_{\text{RL}}(\theta) = -\mathbb{E}_{\mathbf{x} \sim p_{\theta}} [r(\mathbf{x})] + \lambda D_{\text{KL}}(p_{\theta} \| p_{\text{ref}}),$$

where p_{ref} is the SFT checkpoint and λ controls KL regularisation.

Learning or Inference Procedure

Stage 1—SFT. Starting from the Gemma 4 AR checkpoint, the model is fine-tuned on text data using the masked denoising objective above. Less than 10% of the original pre-training token budget is required, as the AR checkpoint provides strong representations that transfer directly to the bidirectional denoising task.

Stage 2—RL + Sampler Distillation. A combined training step jointly (i) uses RL to improve response quality by rewarding high-judge-score outputs and (ii) distils a fast sampler that fills the canvas in fewer denoising steps without quality loss.

Inference. The model is applied with block-autoregressive sampling: a canvas of 256 masked tokens is initialised, then denoised iteratively. Generation can be quantised to 4-bit precision to fit within 18 GB VRAM.

What the Guarantee Says

The two-stage recipe inherits Gemma 4’s general language understanding and instruction following, while the RL stage provides an explicit optimisation target linking generation quality to reward. The KL regularisation in the RL objective bounds the distance from the SFT checkpoint, preventing degenerate solutions that score high on the judge but collapse the output distribution. The sampler distillation guarantee is that the fast sampler matches the full sampler’s output distribution within a KL tolerance tuned during training.

Experimental Findings

- **1000+ tokens/second** on a single NVIDIA H100 (batch size 1, int4 quantisation).
- **700+ tokens/second** on a consumer NVIDIA GeForce RTX 5090.
- **Up to 4× faster** than the Gemma 4 AR baseline on dedicated GPU hardware.
- Fits in **18 GB VRAM** when quantised to 4-bit, enabling local deployment on high-end consumer GPUs

without multi-GPU setup.

- Quality competitive with the equivalent AR model on standard instruction-following and language-understanding benchmarks.

Ablations and Interpretation

Stage 1 only vs. Stage 1+2. The SFT-only checkpoint has higher variance in denoising step count; Stage 2 distillation produces a sampler that converges in fewer iterations, explaining the efficiency gains beyond what the base denoiser achieves.

Canvas size. The 256-token canvas size balances between parallelism (larger is better) and the difficulty of predicting many tokens at once with high confidence (larger is worse). Ablations showed 256 to be near-optimal for the 26B parameter scale.

MoE vs. dense. The 26B-A4B MoE architecture activates fewer parameters per token than a dense 4B model would but leverages a much larger total capacity. The sparse activation is essential for hitting the latency targets at this parameter scale.

Reference

DiffusionGemma Team (Adrien Ali Taïga et al.). **DiffusionGemma Technical Report**. arXiv:2608.00146, August 2026. <https://arxiv.org/abs/2608.00146>